

04 — Product & UX Specification

Product Experience, Interaction Model & UX Behavior Baseline

Field	Value
Document ID	CloudShield AI — 04
Version	1.0
Status	Engineering Baseline
Scope	Product UI/UX and user-facing behavior across CloudShield AI
Primary Cloud	AWS for MVP
Design Principle	Make system state, risk, evidence and safety gates understandable before action.
DAIL Position	Integrated lifecycle/safety subsystem; surfaced through product workflows, not presented as the product.

The interface should make the safe path the obvious path.

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1. Purpose and Scope

This specification defines the product experience of CloudShield AI: how users navigate the platform, understand cloud posture, inspect findings, use AI assistance, review infrastructure changes, observe DAIL lifecycle gates, inspect evidence, and make governed decisions.

The UX must represent the underlying functional model faithfully. The interface must never imply that an AI recommendation is equivalent to a verified or trusted infrastructure state.

1.1 Product UX Boundary

- CloudShield AI is the primary product experience.
- DAIL appears as lifecycle and safety controls attached to remediation/change workflows.
- AI is presented as an assistant/reasoning layer, not an autonomous operator.
- Observed, proposed, verified and trusted states must be visually and semantically distinguishable.
- Evidence and audit history must be reachable from material decisions and state transitions.
- MVP prioritizes clarity and correctness over visual complexity.

2. UX Principles

Principle	UX Requirement
State clarity	Always show whether information is observed, proposed, verified or trusted.
Safety visibility	Blocking gates and failed verification must be prominent and understandable.
Evidence first	Users should be able to inspect why the system reached a conclusion.
AI transparency	AI-generated content must be clearly identified and contextualized.
Progressive disclosure	Show concise summaries first; allow deeper evidence and technical detail on demand.
Actionability	Every major finding should make the next safe action discoverable.
No false certainty	Unknown, stale, partial or failed states must not look healthy or complete.
Least privilege	Controls unavailable to the current role should be hidden or clearly disabled with reason.
Traceability	Users should be able to move from finding →candidate → verification →decision →evidence.
Consistency	State names, severity labels, lifecycle terminology and action patterns remain consistent.

Primary interaction rule: destructive or infrastructure-affecting actions require explicit context, visible scope, preconditions and confirmation.

3. Users, Roles and Jobs-to-be-Done

Role	Primary Jobs	UX Priority
Platform Administrator	Connect cloud accounts, configure scopes/policies, monitor platform health.	Control and operational clarity
Security Analyst	Investigate findings, understand risk, evaluate recommendations.	Evidence and prioritization
Infrastructure Engineer	Inspect resources, review candidate	Technical detail and change safety

	changes, verify technical impact.	
Approver	Review candidate impact/evidence and approve or reject.	Decision clarity
Auditor	Trace historical actions and evidence.	Immutable history and provenance
Automation/Worker	Execute background operations.	Operational status and machine-readable outcomes

3.1 UX by Decision Need

- What is wrong?
- How important is it?
- What resource is affected?
- Why does CloudShield AI believe this?
- What does the AI recommend?
- What exactly would change?
- What could the change affect?
- Has the change been verified?
- Who approved it?
- What is trusted now?
- What changed since the last trusted state?

4. Product Information Architecture

The primary navigation should reflect the product's mental model rather than internal service boundaries.

Primary Area	Purpose	Representative Screens
Overview	Current posture and operational summary.	Dashboard
Cloud	Connections, accounts, regions and discovery.	Connections, Discovery Jobs
Inventory	Assets and configurations.	Assets, Asset Detail, Configuration
Security	Findings and security posture.	Findings, Finding Detail
Risk	Prioritized risk view.	Risk Overview, Risk Detail
Changes	Remediation candidates and lifecycle.	Candidates, Candidate Detail
AI	AI insights, recommendations and evaluation context.	AI Insight Panel, AI Activity
Evidence	Audit, verification and provenance.	Evidence Explorer, Audit Timeline
Operations	Jobs, integrations, health and system events.	Jobs, Integrations, System Health
Settings	Policies, users/roles and product configuration.	Settings

4.1 Navigation Rule

The UI should avoid forcing users to understand DAIL's internal architecture. Users should encounter DAIL through meaningful product concepts such as candidate lifecycle, impact, verification, approval and trusted state.

5. Global Navigation and Application Shell

- Persistent left navigation or equivalent primary navigation.
- Global tenant/account context selector.
- Global search.

- Current environment/scope indicator.
- Notifications/status entry point.
- User identity and role context.
- Breadcrumbs on deep technical pages.
- Consistent page title, scope, freshness and primary action region.

UX-SHELL-001: Every page shall make the active cloud/account scope clear.

UX-SHELL-002: Every data-heavy page shall show data freshness where stale data could affect a decision.

UX-SHELL-003: Primary actions shall be visually distinct from secondary/navigation actions.

UX-SHELL-004: The UI shall not expose an action as executable when required authorization or lifecycle preconditions are not satisfied.

6. Core Dashboard

The dashboard is the operational starting point, not a generic analytics page.

6.1 Dashboard Modules

- Cloud connection health.
- Inventory coverage and discovery freshness.
- Security posture summary.
- Top risk findings/assets.
- Open remediation candidates by lifecycle state.
- Verification failures and blocked changes.
- Recent drift events.
- Recent trusted-state changes.
- Active/failed background jobs.
- Evidence/audit activity summary.

6.2 Dashboard Interaction

- Cards should drill into the filtered underlying dataset.
- Counts should expose scope and timestamp.
- Critical items should link directly to actionable detail.
- The dashboard should distinguish current observed posture from trusted posture where relevant.
- Partial discovery or stale data should be visible rather than hidden behind normal-looking counts.

7. Cloud Connections Experience

7.1 Connection List

- Connection name and account identifier.
- Connection status.
- Last successful validation.
- Last discovery.
- Scope/regions.
- Coverage or permission warnings.
- Actions available to the current role.

7.2 Connection Setup

30. Enter connection metadata and selected scope.
31. Provide/configure supported credential mechanism.
32. Validate access.
33. Show required/optional permission results.
34. Allow the user to correct failures.
35. Mark the connection READY only after required validation succeeds.

UX-CONN-001: Credential input must never expose secrets after submission.

UX-CONN-002: Validation results shall identify missing permissions without displaying sensitive credential material.

UX-CONN-003: A connection with incomplete scope shall visibly communicate reduced coverage.

8. Inventory and Asset Experience

8.1 Asset List

Column/Facet	Purpose
Asset	Human-readable resource identity.
Type	Canonical resource type.
Account/Region	Cloud scope.
Status	Current lifecycle/observation status.
Risk	Current risk priority.
Findings	Number/severity of associated findings.
Last Observed	Freshness indicator.
Trust	Relevant trusted-state status where applicable.

8.2 Asset Detail

- Identity and provider identifiers.
- Current observed configuration.
- Trusted configuration/state.
- Configuration history and drift.
- Security findings.
- Risk context.
- Dependencies and dependents.
- Related candidates.
- Evidence timeline.

Technical JSON/provider details should be available through an expandable/raw-data view rather than overwhelming the primary page.

9. Security Findings Experience

9.1 Findings List

- Severity and risk priority.
- Finding title and concise explanation.
- Affected asset.

- Exposure/context indicators.
- Status.
- Last observed.
- Remediation availability.
- Suppression status where applicable.

9.2 Finding Detail

36. State the problem in plain language.
37. Show the affected resource and scope.
38. Show evidence supporting the finding.
39. Explain why it matters and how risk is calculated.
40. Show dependencies/impact context.
41. Show AI explanation/recommendation if available.
42. Offer safe remediation path.
43. Show candidate and verification history if remediation exists.

UX-FIND-001: A finding shall not present AI-generated text as authoritative evidence.

UX-FIND-002: A finding shall provide a visible path to the underlying evidence.

UX-FIND-003: Suppressed findings shall visibly indicate suppression and reason to authorized users.

10. Risk and Prioritization Experience

Risk UX must answer 'why this is high priority' rather than merely display a number.

- Risk score/level and score version.
- Primary contributing factors.
- Affected asset criticality.
- Exposure.
- Dependency/blast-radius context.
- Confidence/evidence quality.
- Compensating controls where modeled.
- Recommended investigation or remediation path.

UX-RISK-001: Risk score explanations shall identify major contributing factors.

UX-RISK-002: The UI shall distinguish risk from finding severity.

11. Dependency and Impact Experience

Graph visualization is useful but must not be the only representation of impact.

- Summary of directly affected assets.
- Summary of transitively affected assets.
- Critical dependents.
- Potentially invalidated findings.
- Potentially invalidated verification/trusted assumptions.
- Impact confidence/limitations.
- Expandable dependency graph.

- Tabular/list fallback for accessibility and precision.

UX-IMPACT-001: Before a consequential change is approved, the UI shall expose its target scope and known impact.

UX-IMPACT-002: Unknown or incomplete dependency information shall be communicated explicitly.

12. AI Intelligence Experience

AI is presented as a reasoning assistant. The UI must make the distinction between observed evidence and generated interpretation obvious.

12.1 AI Insight Panel

- What the AI was asked to analyze.
- Context used at a high level.
- Explanation.
- Recommendation.
- Assumptions/uncertainties.
- Confidence or evaluation metadata where supported.
- Model/provider metadata appropriate to the user's role.
- Link to source evidence.
- Action to create a candidate, if the recommendation is actionable.

12.2 AI Interaction Rules

UX-AI-001: AI-generated content shall carry an explicit AI-generated indicator.

UX-AI-002: The UI shall not label an AI recommendation as verified, approved or trusted.

UX-AI-003: Users shall be able to inspect the evidence/context supporting an AI recommendation at an appropriate level of detail.

UX-AI-004: Actionable AI recommendations shall transition into the explicit candidate workflow.

UX-AI-005: If AI output is unavailable or invalid, the product shall provide a deterministic fallback/error state rather than implying success.

13. Remediation Candidate Experience

13.1 Candidate List

- Candidate identity.
- Origin.
- Affected resource(s).
- Current lifecycle state.
- Risk/finding linkage.
- Impact status.
- Verification status.
- Approval status.
- Last updated.

13.2 Candidate Detail

44. Show the candidate's purpose.
45. Show exact intended changes.
46. Show current observed state.
47. Show expected post-change state.
48. Show affected resources.
49. Show DAIL lifecycle/technical gates.
50. Show impact analysis.
51. Show verification results.
52. Show governance decision.
53. Show promotion/trusted-state result.
54. Show complete evidence timeline.

UX-CAND-001: The exact candidate version under review shall be unambiguous.

UX-CAND-002: The primary action shall reflect the candidate's current allowed transition.

UX-CAND-003: A blocked candidate shall explain the blocking condition and required next step.

UX-CAND-004: Promotion controls shall remain unavailable until all required gates pass.

14. DAIL Lifecycle Experience

DAIL should be represented through a clear lifecycle stepper/timeline rather than requiring users to understand its internals.

Lifecycle Step	User-Facing Label	Typical Output
Identity	Change Identity	Candidate/version identity
Dependencies	Dependencies	Affected relationships
Impact	Impact Analysis	Blast radius/invalidation
Verification	Verification	Checks and result
Governance	Approval	Decision
Promotion	Promotion	Trusted-state transition

UX-DAIL-001: The DAIL lifecycle view shall expose the current gate and completed gates.

UX-DAIL-002: The UI shall distinguish 'not run', 'running', 'passed', 'failed' and 'blocked' gate states.

UX-DAIL-003: DAIL technical details shall be available for engineers without making them mandatory for ordinary users.

15. Verification and Promotion Experience

15.1 Verification View

- Candidate identity/version.
- Verification checks.
- Expected result.
- Observed result.
- Pass/fail status.
- Evidence.

- Failure explanation.
- Timestamp and execution identity.

15.2 Approval View

- What will change.
- Why it is being changed.
- What could be affected.
- Verification status.
- Policy requirements.
- Evidence.
- Approval/rejection controls.
- Required decision reason.

15.3 Promotion Confirmation

- Restate exact scope.
- Restate verification result.
- Restate governance result.
- Explain that promotion changes trusted state.
- Require explicit confirmation when policy requires it.
- Show resulting trusted-state update and evidence.

UX-PROM-001: Promotion confirmation shall clearly state that trusted state will change.

UX-PROM-002: A failed precondition shall disable promotion and explain the reason.

UX-PROM-003: The UI shall not report a promotion as successful until backend confirmation is received.

16. Trusted State and Drift Experience

Trusted state should be visible as a distinct concept. Users must be able to understand whether current cloud state matches the state that CloudShield AI has accepted.

Indicator	Meaning
Observed	Latest provider-reported state.
Matches Trusted	Observed state matches trusted state for the modeled scope.
Drifted	Observed state differs from trusted state.
Pending Change	A candidate is expected to alter the state.
Verification Required	The state cannot yet be treated as verified.
Unknown/Stale	The system lacks sufficiently fresh evidence.

UX-TRUST-001: Trusted state shall never be represented solely by a generic 'healthy' indicator.

UX-TRUST-002: Drift views shall show the difference between observed and trusted state.

17. Evidence, Audit and Reporting Experience

17.1 Evidence Timeline

- Chronological material events.

- Actor/service identity.
- Event type.
- Related object.
- Timestamp.
- Evidence/artifact link.
- Decision rationale where applicable.

17.2 Reporting

- Current posture summary.
- Open findings and risk.
- Remediation progress.
- Verification outcomes.
- Drift/trusted-state changes.
- Cloud coverage and discovery freshness.
- Audit/export capabilities subject to authorization.

UX-EVID-001: Material state transitions shall be traceable from the UI to their evidence.

UX-EVID-002: Audit history shall distinguish human decisions from automated/system actions.

18. Jobs, Notifications and System Status

18.1 Job Center

- Job type.
- Scope.
- Status.
- Progress where meaningful.
- Started/updated/completed timestamps.
- Worker/execution identity where appropriate.
- Error/partial-failure summary.
- Links to affected results.

18.2 Notifications

- Critical finding detected.
- Discovery completed/failed/partial.
- Verification failed.
- Candidate blocked.
- Approval required.
- Promotion completed/failed.
- Material drift detected.
- Cloud connection degraded.

UX-JOB-001: Notifications shall link directly to the relevant object or job detail.

UX-JOB-002: The UI shall not represent a queued/running job as completed.

19. Search, Filtering and Data Exploration

- Global search should find assets, findings, candidates, jobs and evidence by supported identifiers.
- Lists should support filters appropriate to the object type.
- Filters should be visible and removable.
- Deep links should preserve relevant filter/scope context where practical.
- Large datasets should support pagination or equivalent bounded retrieval.
- Sorting should be deterministic and consistent.

Object	High-Value Filters
Assets	Type, account, region, risk, finding count, freshness, trust/drift
Findings	Severity, risk, state, asset type, account, exposure, suppression
Candidates	State, origin, risk, verification, approval, account
Jobs	Type, status, scope, date, failure state
Evidence	Object, event type, actor, date, outcome

20. Forms, Validation and Confirmation Patterns

- Validate required fields before submission.
- Use inline validation for recoverable input errors.
- Never clear valid user input after an unrelated validation error.
- Explain why a field is required when the reason is not obvious.
- For consequential actions, present a review step before execution.
- Confirmation dialogs must describe scope, consequence and irreversible/long-running behavior.
- Use idempotency-safe UX for retried actions rather than encouraging duplicate submission.

UX-FORM-001: Forms shall distinguish validation errors from authorization and backend execution failures.

UX-FORM-002: Consequential actions shall require explicit confirmation where defined by policy.

21. Error, Empty and Partial-State UX

State	UX Treatment
Loading	Show progress appropriate to expected latency; avoid fake completion.
Empty	Explain why no data exists and provide a useful next action.
Stale	Show last-known timestamp and freshness warning.
Partial	Identify missing scope/results and what failed.
Permission denied	Explain lack of access without exposing sensitive details.
AI unavailable	Show AI unavailable state and preserve deterministic product functionality.
Verification failed	Highlight failure and block unsafe progression.
Promotion failed	Clearly state trusted state was not advanced unless backend confirms otherwise.
Unknown	Use explicit unknown state rather than defaulting to healthy/pass.

UX-ERR-001: The product shall never substitute a visually healthy state for unknown or failed data.

UX-ERR-002: Partial results shall be labeled as partial wherever completeness affects interpretation.

22. Accessibility and Usability Requirements

- Core workflows must be keyboard navigable.
- Status should not rely solely on color.
- Interactive controls require accessible names.
- Tables and dense technical content must remain readable at increased text size.
- Graph views require a non-graph alternative.
- Focus order must follow the logical task sequence.
- Error messages must be programmatically associated with relevant inputs.
- Critical safety states must have text labels in addition to visual indicators.

UX-A11Y-001: All primary MVP workflows shall be usable without pointer-only interaction.

UX-A11Y-002: Lifecycle and risk status shall have text semantics independent of color.

UX-A11Y-003: Dependency/impact information shall have a list/table representation.

23. Security and Privacy UX

- Never display cloud credentials, tokens or secrets in normal UI.
- Mask sensitive identifiers where full values are not required.
- Clearly identify tenant/account scope before actions.
- Show role/permission limitations when an action is unavailable.
- Audit sensitive administrative actions.
- Avoid placing sensitive infrastructure details in notifications where not required.
- Use explicit session/authentication state indicators.

UX-SEC-001: Secret material shall never be rendered as ordinary application content.

UX-SEC-002: Users shall see the scope affected by a consequential action before confirmation.

24. Responsive and Performance UX

The primary target is a desktop engineering/security console. Responsive behavior should preserve usability on smaller screens without compromising critical state and evidence.

- Primary desktop workflows are optimized for dense technical information.
- Long-running operations are asynchronous with visible job status.
- Pages should progressively load secondary evidence/details where appropriate.
- Large tables should avoid rendering unbounded records.
- The UI should remain usable when AI or a cloud provider is temporarily unavailable.
- Expensive graphs should support bounded/filtered views.

UX-PERF-001: User actions that start long-running operations shall return control to the user promptly and expose job progress.

UX-PERF-002: The UI shall not require a synchronous browser wait for discovery, verification or other long-running backend workflows.

25. MVP Screen Set and User Flows

25.1 MVP Screens

Screen	Priority	Purpose
Dashboard	P0	Current posture and operational overview.
Cloud Connections	P0	Configure and validate AWS connections.
Discovery Jobs	P0	Run and inspect discovery.
Asset List	P0	Explore normalized inventory.
Asset Detail	P0	Inspect configuration, risk, findings and dependencies.
Findings	P0	Prioritize security conditions.
Finding Detail	P0	Investigate and initiate remediation.
Candidate List	P0	Track remediation lifecycle.
Candidate Detail	P0	Review exact change, impact, DAIL gates and evidence.
Verification	P0	Inspect deterministic verification.
Approval	P0	Govern candidate promotion.
Evidence Timeline	P0	Trace decisions and state transitions.
Jobs	P0	Monitor asynchronous operations.
Settings	P1	Policies, roles and configuration.

25.2 Primary Golden Path

55. Connect AWS account.
56. Validate connection.
57. Run discovery.
58. Review dashboard and findings.
59. Open a high-priority finding.
60. Inspect evidence and AI explanation.
61. Create remediation candidate.
62. Review impact and DAIL lifecycle.
63. Run verification.
64. Approve candidate.
65. Promote after all gates pass.
66. Observe trusted-state update.
67. Return to finding and confirm resolution/evidence.

25.3 Secondary Path — Drift

68. Monitoring runs discovery.
69. Observed state differs from trusted state.
70. Dashboard shows drift.
71. User opens affected asset.
72. Configuration diff identifies changed fields.
73. Relevant findings/risk are refreshed.
74. User may create a new candidate if remediation is appropriate.

26. UX Acceptance Criteria

UX-ACC-01: Every core page exposes active scope and relevant freshness.

- UX-ACC-02:** Observed, proposed, verified and trusted states are distinguishable.
- UX-ACC-03:** Dashboard provides direct navigation to high-priority security and lifecycle issues.
- UX-ACC-04:** Finding detail exposes evidence, risk context and safe next actions.
- UX-ACC-05:** AI content is visibly identified and cannot be mistaken for verified state.
- UX-ACC-06:** AI recommendations flow into explicit candidates.
- UX-ACC-07:** Candidate detail exposes exact change scope and expected state.
- UX-ACC-08:** Impact analysis is visible before consequential approval.
- UX-ACC-09:** DAIL lifecycle gates are understandable without requiring internal DAIL knowledge.
- UX-ACC-10:** Verification failures are visually prominent and prevent promotion actions.
- UX-ACC-11:** Approval screens present change, impact, verification and evidence before decision.
- UX-ACC-12:** Promotion explicitly communicates trusted-state consequences.
- UX-ACC-13:** Promotion success is only shown after confirmed backend completion.
- UX-ACC-14:** Drift clearly compares observed and trusted state.
- UX-ACC-15:** Evidence can be reached from material findings, candidates and decisions.
- UX-ACC-16:** Partial/stale/unknown states are not displayed as healthy.
- UX-ACC-17:** Long-running jobs expose status and do not block the main UI.
- UX-ACC-18:** Critical workflows are keyboard accessible.
- UX-ACC-19:** Dependency/impact information has a non-graph representation.
- UX-ACC-20:** Secret material is never exposed through normal product UX.

27. Traceability to Functional Specification

UX Area	Functional Specification Alignment
Dashboard	Functional areas, monitoring, reporting, jobs
Cloud connections	Cloud onboarding and authorization
Inventory	Discovery, normalization, configuration snapshots
Findings	Security analysis and finding lifecycle
Risk	Risk intelligence and prioritization
Dependencies/Impact	Dependency and impact analysis
AI	AI intelligence workflow and abstraction
Candidates	Remediation and candidate lifecycle
DAIL	DAIL integration workflow
Verification	Verification and promotion
Trusted state	Trusted state and monitoring
Evidence	Evidence, audit and reporting
Jobs	Asynchronous processing
Errors	Error and exception behavior

The UX specification does not redefine backend semantics. It defines how those semantics are exposed to users, how ambiguity is communicated, and how safety-critical decisions are presented.

28. Future UX Evolution

- Multi-cloud views using the same normalized product concepts.
- Advanced topology visualization.
- Policy authoring and simulation.
- Change-plan comparison and richer infrastructure diffs.
- AI conversational investigation with controlled context access.
- Local/open-weight model indicators and model-selection controls where appropriate.
- Advanced risk dashboards and historical trend analysis.
- Custom reports and organization-specific dashboards.
- Collaborative review and annotation workflows.
- Mobile/compact operational views for alerts and approvals.

Future features must preserve the same safety semantics: AI remains advisory, changes are explicit candidates, verification is deterministic, governance is explicit, evidence is retained, and trusted state is earned.

29. Glossary

Term	UX Meaning
Observed	Latest provider-reported state.
Proposed	State a candidate intends to create.
Verified	State confirmed by deterministic verification.
Trusted	State accepted after required governance/promotion.
Finding	Security/configuration condition identified by analysis.
Risk	Prioritization signal based on finding and context.
Candidate	Explicit infrastructure change representation.
DAIL	Deterministic AI Infrastructure Lifecycle subsystem.
Impact	Resources and assumptions potentially affected by a candidate.
Evidence	Artifacts supporting system observations, decisions and transitions.
Drift	Difference between relevant observed and trusted state.

Document Completion Statement

CloudShield AI Product & UX Specification v1.0 establishes the user-facing interaction contract for the product. It is designed around the complete CloudShield AI control loop and intentionally exposes DAIL as a safety/lifecycle capability within the product rather than presenting DAIL as the product itself.

Next planned document: 05 — CloudShield AI Domain & Data Model Specification.